

Milestone 1

Software Engineering Project AI-powered Pharmacy Management System Team 65

Identification of Users

Based on the organizational structure, the system will cater to three distinct user tiers:

Primary Users

Direct users of the application.

Staff Members: Managing day-to-day inventory, processing billing, and restocking alerts.

Patients: Engaging with the chat interface, booking appointments, and submitting prescriptions.

Secondary Users

Indirect users of the application.

Admin: Overseeing high-level analytics and managing registered doctor schedules.

Tertiary Users

Does not use the application at all, but is affected by it.

Registered Doctors: External practitioners receiving appointments via the platform.

User Research Interviews

The following section documents the pain points uncovered through user interviews and observation studies conducted with the pharmacy owner and customers. Interviewees were NOT asked whether specific features would be useful; the purpose was to understand real problems they face in daily operations.

Audio recordings of all interviews are available at the following link:

Team 65 Interview Recordings: [📺 Interviews](#)

Interview 1: Shop Owner (Business Perspective)

Interviewee: Mr. Saha, Owner, Saha Pharmacy. Method: In-person. The recording has been uploaded to the Team 65 shared drive linked above.

Key Pain Points Identified

Source	Pain Point Identified
Mr. Saha (Owner)	No digital inventory management system - stock is checked visually and noted in a paper notebook. There is no formal tracking process.
Mr. Saha (Owner)	Stockouts occur during busy hours without prior warning. The owner only notices when the shelf is already empty, leading to customer dissatisfaction.
Mr. Saha (Owner)	Hand-written bills are produced using a calculator during rush hours. Rechecking is required when discounts are applied or when there are many items.
Mr. Saha (Owner)	End-of-month account reconciliation is entirely manual, collecting all paper bills and recalculating totals. Mismatches require re-checking everything, consuming significant time.
Mr. Saha (Owner)	Doctors' handwriting is difficult to read, particularly for new staff members. Unclear prescriptions require calling the doctor to confirm, which takes additional time.
Mr. Saha (Owner)	Customers regularly ask about medicine usage, side effects, and cheaper alternatives. Answering during busy hours creates operational pressure.
Mr. Saha (Owner)	No structured tracking of which medicines sell the most. Exact monthly figures cannot be provided on the spot when asked by suppliers or stakeholders.
Mr. Saha (Owner)	Supplier queries about last month's purchase quantities require manual notebook searching and cannot be answered instantly.
Mr. Saha (Owner)	Expiry date checking is manual. Medicines sometimes remain on shelves past their near-expiry window, resulting in financial loss when they cannot be sold.
Mr. Saha (Owner)	Doctor appointments are managed only via phone calls or walk-in requests with no fixed schedule. Double-bookings and conflicts arise, disturbing regular shop operations.

Interview 2: Customer Interaction 1 - Sreejita Chatterjee

Interviewer: Piyush Saha. Method: Phone call. Sreejita Chatterjee is a regular customer residing in Dum Dum, near Saha Pharmacy.

Key Pain Points Identified

Source	Pain Point Identified
Sreejita Chatterjee (Patient)	Waiting time during evening rush hours is 15 to 20 minutes. Standing in a queue while unwell is physically uncomfortable and frustrating.
Sreejita Chatterjee (Patient)	Cannot read the doctor's prescription herself; she is fully dependent on the pharmacist to identify and dispense the correct medicines.
Sreejita Chatterjee (Patient)	Feels hesitant to ask follow-up questions about her medicines because people are waiting behind her, leaving her with unresolved doubts.
Sreejita Chatterjee (Patient)	Medicines are occasionally out of stock, requiring her to visit another pharmacy. This is especially inconvenient when the medicine is urgently needed.

Sreejita Chatterjee (Patient)	Manual billing during rush hours is slow. Even small delays feel significant when one is tired or unwell after a long day at work.
--------------------------------------	--

Interview 3: Customer Interaction 2 - Soumyadipta Das

Interviewer: Piyush Saha. Method: Phone call. Soumyadipta Das is a regular customer in Dum Dum who purchases chronic-condition medications (blood pressure, diabetes) monthly.

Key Pain Points Identified

Source	Pain Point Identified
Soumyadipta Das (Patient)	Cannot read prescription clearly and frequently forgets exact medicine names; entirely dependent on the pharmacist for accurate dispensing every month.
Soumyadipta Das (Patient)	At an older age, standing continuously during long queues is physically uncomfortable. A faster service process is desired.
Soumyadipta Das (Patient)	Has received medicine under a different brand name from what was prescribed and was unsure whether it was the same medicine, causing anxiety about his health and well-being.
Soumyadipta Das (Patient)	Manual billing and calculation by hand is slow, especially during busy hours when many customers are present simultaneously.

Interview 4: Customer Interaction 3 - Barnik Chakraborty

Interviewer: Piyush Saha. Method: Phone call. Barnik Chakraborty resides in Dum Dum and visits the pharmacy primarily for his nephew's urgent medical needs.

Key Pain Points Identified

Source	Pain Point Identified
Barnik Chakraborty (Patient)	Visits are typically urgent (child's illness). Crowded shop with long waiting times cause significant anxiety when a child needs medication quickly.
Barnik Chakraborty (Patient)	Medicine names are complex and difficult to remember. Has to double-check the name of the dispensed medicine himself due to fear of receiving the wrong one.
Barnik Chakraborty (Patient)	No systematic accuracy verification is done at the counter; the process is entirely manual, raising concern about dispensing errors for a child's medication.
Barnik Chakraborty (Patient)	A prescribed syrup was once out of stock, requiring him to visit another pharmacy with an unwell child - a stressful and tiring experience.
Barnik Chakraborty (Patient)	Dosage instructions (frequency, before/after meals) must be verbally asked during the busy visit. He still feels uncertain about the dosage after returning home.
Barnik Chakraborty (Patient)	Manual billing causes delay. When a child is unwell at home and waiting, even small delays at the pharmacy counter feel considerably longer.

Consolidated Pain Points Summary

Across all four interviews, the following core pain points were consistently identified:

- No digital stock management - all inventory tracking is manual and error-prone.
- Stockouts without early warning - customers either leave unsatisfied or must visit competitor pharmacies.
- Slow manual billing - causes long queues, delays, and frustration, especially for unwell patients.
- Difficulty reading handwritten prescriptions - affects both new staff efficiency and patient safety.
- No structured medicine sales data - prevents informed restocking and supplier communication.
- Expiry date management is reactive and manual - leading to financial losses from expired stock.
- Doctor appointment scheduling has no system - results in double bookings and workflow disruptions.
- Patients cannot verify medicine details independently - reliance on pharmacist creates bottleneck.
- No mechanism for patients to check medicine availability before visiting the store.
- Dosage and usage instructions are communicated verbally only - patients leave with residual doubts.

SMART User Stories

The following user stories are written based on pain points identified through interviews. Each story follows the SMART framework: Specific, Measurable, Achievable, Relevant, and Time-bound.

User Management

Story: As a staff member, I want to login to the system, so that I can quickly create bills and search for inventory levels for products.

Specific	Login authentication system for pharmacy.
Measurable	Should take less than 2 minutes for user registration.
Achievable	Viable within one week of development.
Relevant	Only give access to staff members.
Time-bound	This will be completed within the first two weeks once development begins.

Inventory Management System

Story: As a staff user, I want to know the stock of a particular medicine, so that I can complete the order in under a minute.

Specific	Query inventory database for stock information.
Measurable	Order takes under a minute.
Achievable	Feasible within two weeks of development.
Relevant	Decreases order time and improves product tracking.

Time-bound	This will be completed within four weeks once development begins.
-------------------	---

Billing System

Story: As a cashier, I want the system to automatically generate a tax-compliant invoice so that I can provide the customer with a record of their purchase.

Specific	Auto-calculate totals, GST/VAT, and discounts to generate a downloadable PDF.
Measurable	Invoice generation takes less than 2 seconds after payment confirmation.
Achievable	Uses existing database hooks; manageable for the backend team.
Relevant	Ensures legal compliance and provides professional documentation.
Time-bound	This will be deployment ready within six weeks once development begins.

Prescription Reading System

Story: As a pharmacist, I want to upload a photo of a handwritten prescription so that the system can automatically extract the drug names and dosages.

Specific	Implement an OCR using LLM to parse medical text from images.
Measurable	Achieve at least 85% accuracy in text extraction.
Achievable	High complexity; requires using a pre-trained ML model or API.
Relevant	Drastically reduces manual data entry time and potential human error.
Time-bound	A beta version will be completed within eight weeks once development begins.

Medicine Recommendation System

Story: As a pharmacist, I want the system to suggest generic alternatives for expensive brands so that I can offer more affordable or available options to patients.

Specific	Create a logic-based engine that matches active ingredients (APIs) across the inventory.
Measurable	Return at least two generic alternatives for any "Brand Name" query (if available).
Achievable	Relies on a well-structured medicine database.
Relevant	Improves customer satisfaction and increases inventory turnover.
Time-bound	This feature will be live within seven weeks once development begins.

Data Analytics Dashboard

Story: As a pharmacy manager, I want to view a visual summary of monthly sales and top-selling medicines so that I can make informed restocking decisions.

Specific	Develop a dashboard with bar charts and line graphs showing sales trends.
Measurable	Dashboard must refresh with 100% accurate data upon page load.
Achievable	Feasible using a frontend charting library (like Chart.js or D3).
Relevant	Critical for strategic planning and financial oversight.
Time-bound	This will be finalized and polished within nine weeks once development begins.

Stock Reminder System

Story: As a store owner, I want to receive an automated alert when any medicine falls below its minimum threshold so that I never run out of essential stock.

Specific	Setup an automated email/SMS notification trigger based on real-time inventory counts.
Measurable	Notification is sent within 10 minutes of the stock hitting the "Low" threshold.
Achievable	Simple trigger logic within the existing database.
Relevant	Prevents lost sales and ensures patients have access to life-saving meds.
Time-bound	This will be implemented within five weeks once development begins.

Appointment Booking System

Story: As a patient, I want to book a consultation with a pharmacist online so that I can avoid waiting in a physical queue.

Specific	A calendar-based interface for selecting available 15-minute time slots.
Measurable	The entire booking process must be completed in 4 clicks or fewer.
Achievable	Standard CRUD operations; well within the team's capabilities.
Relevant	Optimizes staff time and improves the overall patient experience.
Time-bound	This will be ready for UAT (User Acceptance Testing) within ten weeks once development begins.

Expiry Date Management System

Story: As a store owner, I want the system to automatically flag medicines that are approaching their expiry date, so that I can take proactive action and prevent financial losses from expired stock.

Specific	Implement a date-based alert system that flags medicines within 30 days of expiry.
Measurable	Alert generated at least 30 days before the expiry date; zero expired medicine losses within 2 months of deployment.

Achievable	Date-comparison trigger logic on the inventory database; feasible within two weeks of development.
Relevant	Directly addresses financial losses caused by expired medicines identified in the owner interview.
Time-bound	This will be implemented and live within six weeks once development begins.

Medicine Availability Check for Patients

Story: As a patient, I want to check whether a specific medicine is available at the pharmacy before visiting, so that I can avoid wasted trips when the medicine is out of stock.

Specific	Expose a patient-facing search interface that queries real-time inventory availability by medicine name.
Measurable	Search result returned in under 3 seconds; medicine availability status is 100% accurate relative to live stock levels.
Achievable	Read-only query on the existing inventory database; achievable within one week once the inventory module is complete.
Relevant	Addresses a recurring pain point across all three customer interviews -- out-of-stock situations forcing patients to visit competitor pharmacies.
Time-bound	This will be available to patients within seven weeks once development begins.

Dosage and Usage Information System

Story: As a patient, I want to receive a printed or digital dosage instruction summary along with my purchased medicines, so that I can refer to it at home without relying solely on verbal communication at the counter.

Specific	Auto-generate a dosage instruction slip (frequency, timing, and warnings) for each dispensed medicine, printed or sent digitally at checkout.
Measurable	Instruction slip generated for 100% of dispensed items; generation adds no more than 5 seconds to billing time.
Achievable	Template-based generation tied to the medicine database and billing workflow; achievable within two weeks of development.
Relevant	Addresses residual patient uncertainty about dosages found in customer interviews, particularly for elderly patients and parents managing children's health.
Time-bound	This will be deployed alongside the billing system within six weeks once development begins.

Supplier Purchase History Reporting

Story: As a store owner, I want to instantly generate a report of purchase quantities per medicine for any given month, so that I can respond to supplier queries without manually searching through paper notebooks.

Specific	Build a report generation feature that filters purchase records by medicine and date range, exportable as PDF or CSV.
Measurable	Report generated in under 5 seconds for any selected month; data accuracy verified against billing records.
Achievable	Standard database query with date-filter and export; feasible within one week after the billing module is complete.
Relevant	Directly resolves the owner's inability to respond to supplier queries instantly, as identified in the owner interview.
Time-bound	This will be available within nine weeks once development begins, alongside the analytics dashboard.